

HUANCONG GUAN

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RESEARCH INTERESTS

3D Computer Vision, 3D Gaussian Splatting, 3D Scene Understanding

EDUCATION

Shenzhen University (SZU) , Shenzhen, China	Sep 2024 – Present
M.Sc. in Computer Science and Technology, College of Artificial Intelligence	GPA: 3.42/4.00
Guangdong University of Technology (GDUT) , Guangzhou, China	Sep 2020 – Jun 2024
B.Eng. in Internet of Things Engineering, School of Automation	GPA: 3.75/5.00

PUBLICATION

Robust Instance Segmentation via 3D Gaussian Denoising Constraints
Huancong Guan, Zehai Niu, Jiayi Lyu, Teng Wang, Bin Huang, Linlin Shen, Jinbao Wang
IEEE Transactions on Image Processing (TIP), Accepted with Minor Revisions (AQE)

RESEARCH EXPERIENCE

Robust Instance Segmentation via 3D Gaussian Denoising Constraints	Sep 2025 – Present
Shenzhen University Advisor: Dr. Jinbao Wang	

- Proposed DenoGS, a global-local denoising framework for robust 3D Gaussian Splatting instance segmentation under noisy and cross-view inconsistent 2D supervision.
- Designed prototype-based confidence estimation and spatial filtering to identify reliable supervision, and extended the framework to language-guided 3D instance retrieval.
- Achieved 0.82 mIoU and 0.89 F1 across 41 challenging scenes, with consistent improvements over strong baselines under sparse-view settings.

Multimodal Classification of Human Oocyte Development	Dec 2025 – Present
Shenzhen University & Shenzhen Maternity and Child Healthcare Hospital	

- Proposed the first multimodal framework for predicting embryo euploidy versus aneuploidy, using a GRU-based video encoder for temporal representation learning and an MLP-based encoder for integrating tabular features.
- Achieved 77% test accuracy on a dataset of approximately 577 samples.

Semantic-Guided Object Editing in 3D Gaussian Splatting	May 2026 – Present
Shenzhen University Advisor: Dr. Jinbao Wang	

- Designed a remove-repair-restore pipeline with Qwen-based inpainting and multi-view reconstruction to achieve consistent 3D scene editing with reduced artifacts.
- Integrated semantic features with local geometric connectivity, color consistency, and multi-view visibility constraints to achieve accurate Gaussian selection while suppressing background and surface contamination.

AWARDS AND HONORS

Special Graduate Scholarship , Shenzhen University	2024
Second-Class University Scholarship , Guangdong University of Technology	2021, 2023
Second Prize , China Undergraduate Mathematical Contest in Modeling, Guangdong Province	2022
Third Prize , China Electrical Engineering Mathematical Modeling Competition	2022

SKILLS

AI Systems: Large Language Models (LLMs), Agent Frameworks, Retrieval-Augmented Generation (RAG)
Programming Languages: Python, C++
Deep Learning: PyTorch, CNN, Transformer, GRU
Languages: Chinese (Native), English (CET-4, CET-6)